

Practical Week – 11

Task 1 - Develop a multithreaded counter application using POSIX threads (pthread_create, pthread_join). Demonstrate race conditions by allowing multiple threads to update a shared variable concurrently.

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

#define NUM_THREADS 4
#define ITERATIONS 1000000

int counter = 0;

void *increment_counter(void *arg)
{
    for (int i = 0; i < ITERATIONS; i++)
    {
        counter++;
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    printf("Initial counter: %d\n", counter);

    /* Create threads */
    for (int i = 0; i < NUM_THREADS; i++)
    {
        if (pthread_create(&threads[i], NULL,
                          increment_counter, NULL) != 0)
        {
            perror("pthread_create");
            return 1;
        }
    }

    /* Wait for all threads */
    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    printf("Final counter: %d\n", counter);

    printf("Expected counter: %d\n",
           NUM_THREADS * ITERATIONS);

    return 0;
}
```

```

mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap$ mkdir task2_mmap
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap$ cd task2_mmap
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ echo "Hello! This is a file for mma
p demonstration." > input.txt
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ cat input.txt
Hello! This is a file for mmap demonstration.
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ nano mmap_program.c

mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ gcc mmap_program.c -o mmap_program
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ ls
input.txt  mmap_program  mmap_program.c
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ ./mmap_program
File size: 46 bytes

File contents using mmap():
Hello! This is a file for mmap demonstration.

File successfully modified using mmap().
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ cat input.txt
File modified using mmap().map demonstration.
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ nano traditional_program.c
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ gcc traditional_program.c -o tradit
ional_program
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ ./traditional_program
File contents using read():
File modified using mmap().map demonstration.
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ dd if=/dev/zero of=largefile.txt bs
=1M count=100
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 0.0888376 s, 1.2 GB/s
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ cp traditional_program.c traditiona
l_large.c
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ nano traditional_large.c
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ gcc traditional_large.c -o traditio
nal_large
traditional_large.c:3:10: fatal error: stdlib.h: No such file or directory
   3 | #include <stdlib.h>
     |          ^
compilation terminated.
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ cp mmap_program.c mmap_large.c
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ nano mmap_large.c
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ gcc mmap_large.c -o mmap_large
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ time ./mmap_large > mmap_output.txt

real    0m0.717s
user    0m0.032s
sys     0m0.684s
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ mkdir task1_threads
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap$ cd task1_threads

```

```

mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ pwd
/home/mohana_priya/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ nano race_condition.c

mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ gcc race_condition.c
-o race_condition -pthread
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ ls
race_condition  race_condition.c
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ ./race_condition
Initial counter: 0
Final counter: 1258948
Expected counter: 4000000
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ ./race_condition
Initial counter: 0
Final counter: 1088357
Expected counter: 4000000
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ ./race_condition
Initial counter: 0
Final counter: 1451819
Expected counter: 4000000
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ ./race_condition
Initial counter: 0
Final counter: 1084691
Expected counter: 4000000
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ pthread_create(&threads[i], NULL,
    increment_counter, NULL);
-bash: syntax error near unexpected token `&'
-bash: syntax error near unexpected token `)'
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ pthread_join(threads[
i], NULL);
-bash: syntax error near unexpected token `threads[i],'
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$ for i in {1..5}; do .
/race_condition; done
Initial counter: 0
Final counter: 1327331
Expected counter: 4000000
Initial counter: 0
Final counter: 1396840
Expected counter: 4000000
Initial counter: 0
Final counter: 1564791
Expected counter: 4000000
Initial counter: 0
Final counter: 1344581
Expected counter: 4000000
Initial counter: 0
Final counter: 1139172
Expected counter: 4000000
mohana_priya@AMMU:~/task30/task31/task_inode/task_mmap/task2_mmap/task1_threads$

```

Task 2 - Modify the previous program using mutex locks (pthread_mutex_lock, pthread_mutex_unlock) and compare the outputs with the race-condition version. Analyze the impact of synchronization.

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>

#define NUM_THREADS 4
#define ITERATIONS 1000000

int counter = 0;

/* Create mutex */
pthread_mutex_t lock;

void *increment_counter(void *arg)
{
    for (int i = 0; i < ITERATIONS; i++)
    {
        /* Lock before accessing shared variable */
        pthread_mutex_lock(&lock);

        counter++;

        /* Unlock after accessing shared variable */
        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    /* Initialize mutex */
    pthread_mutex_init(&lock, NULL);

    printf("Initial counter: %d\n", counter);

    /* Create threads */
    for (int i = 0; i < NUM_THREADS; i++)
    {
        if (pthread_create(&threads[i], NULL,
                          increment_counter, NULL) != 0)
        {
            perror("pthread_create");
            return 1;
        }
    }

    /* Wait for all threads */
    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    printf("Final counter: %d\n", counter);

    printf("Expected counter: %d\n",
           NUM_THREADS * ITERATIONS);

    /* Destroy mutex */
    pthread_mutex_destroy(&lock);

    return 0;
}
```

```
mohana_priya@AMMU: ~  
mohana_priya@AMMU:~$ cd task1_threads  
-bash: cd: task1_threads: No such file or directory  
mohana_priya@AMMU:~$ ls  
fork_process      memory_analysis  myShell.c        process_states  
fork_process.c    memory_analysis.c practical6         process_states.c  
linux_io_project  myShell          practical_memory  task30  
mohana_priya@AMMU:~$ nano mutex_counter.c  
mohana_priya@AMMU:~$ gcc mutex_counter.c -o mutex_counter -pthread  
mohana_priya@AMMU:~$ ls  
fork_process      memory_analysis  mutex_counter.c  practical6         process_states.c  
fork_process.c    memory_analysis.c myShell          practical_memory  task30  
linux_io_project  mutex_counter    myShell.c        process_states  
mohana_priya@AMMU:~$ ./mutex_counter  
Initial counter: 0  
Final counter: 4000000  
Expected counter: 4000000  
mohana_priya@AMMU:~$ for i in {1..5}; do ./mutex_counter; done  
Initial counter: 0  
Final counter: 4000000  
Expected counter: 4000000  
Initial counter: 0  
Final counter: 4000000  
Expected counter: 4000000  
Initial counter: 0  
Final counter: 4000000  
Expected counter: 4000000  
Initial counter: 0  
Final counter: 4000000  
Expected counter: 4000000  
Initial counter: 0  
Final counter: 4000000  
Expected counter: 4000000  
Initial counter: 0  
Final counter: 4000000  
Expected counter: 4000000  
mohana_priya@AMMU:~$ ./race_condition  
-bash: ./race_condition: No such file or directory  
mohana_priya@AMMU:~$ ./mutex_counter  
Initial counter: 0  
Final counter: 4000000  
Expected counter: 4000000  
mohana_priya@AMMU:~$ pthread_mutex_lock(&lock);  
-bash: syntax error near unexpected token '&'  
mohana_priya@AMMU:~$ time ./race_condition  
-bash: ./race_condition: No such file or directory  
  
real    0m0.002s  
user    0m0.001s  
sys     0m0.001s  
mohana_priya@AMMU:~$ time ./mutex_counter  
Initial counter: 0  
Final counter: 4000000  
Expected counter: 4000000  
  
real    0m0.207s  
user    0m0.269s  
sys     0m0.367s  
mohana_priya@AMMU:~$ |
```